

Monika Kusumanchi

+91 6281074516 | kusumonika033@gmail.com

[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Blog](#)

PROFESSIONAL SUMMARY

Generative AI Engineer with experience building **production-grade LLM applications, conversational AI systems, RAG pipelines, and voice AI agents**. Skilled in designing scalable AI workflows, low-latency conversational systems, prompt evaluation pipelines, and cloud-based AI infrastructure across **AWS and GCP**. Built AI copilots, call analytics platforms, and real-time voice-enabled AI systems with measurable impact, including reducing inference costs by 60%. **Winner of multiple national-level AI hackathons.**

EDUCATION

Bachelor of Technology – CSE (AI & ML) | 2020 – 2024

Jawaharlal Nehru Technological University Kakinada | CGPA: 8.0

SKILLS

- **Programming:** Python, SQL
- **Generative AI & LLMs:** LangChain, RAG Pipelines, Prompt Engineering, LLM Applications, Tool Calling, Structured Outputs, Prompt Evaluation
- **Conversational & Voice AI:** Conversational AI, Voice Agents, ASR/TTS Pipelines, Dialog State Tracking, Intent Detection, Entity Extraction, Multi-turn Conversations
- **Backend & APIs:** FastAPI, REST APIs
- **Cloud & Infrastructure:** AWS (ECS, Lambda, API Gateway), GCP (Cloud Run, BigQuery), Docker, Vercel
- **Databases:** MongoDB, PostgreSQL, Neo4j, BigQuery
- **AI Frameworks & Tools:** Hugging Face, CrewAI, Agno

WORK EXPERIENCE

Runo — AI Engineer | April 2025 – Present

- Designed and deployed production-grade AI copilots that convert natural language into MongoDB queries using LLM orchestration, RAG pipelines, and structured outputs.
- Developed a scalable AI-powered call analytics platform for extracting intent, sentiment, and conversational insights from sales calls, reducing infrastructure costs by 60%.
- Engineered voice-enabled AI agents using speech-to-text pipelines, conversational memory, and multi-turn interaction handling for SaaS workflows.
- Implemented prompt evaluation pipelines, regression testing, and conversational stress testing to improve LLM reliability and response consistency.
- Built and maintained scalable backend infrastructure using AWS ECS, Lambda, API Gateway, Docker, and serverless deployment frameworks.

Solivar — AI Engineer | Jun 2024 – Mar 2025

- Delivered client-facing conversational AI systems and LLM-powered assistants tailored to enterprise business workflows using CrewAI and Agno.
- Designed retrieval-based AI workflows and prompt orchestration pipelines for multi-step task execution and business process automation.
- Engineered scalable backend services using FastAPI, Docker, PostgreSQL, Neo4j, and BigQuery for production AI applications.

CloudKarya — ML Engineer | Jun 2023 – May 2024

- Developed and deployed machine learning applications for medical analytics, restaurant revenue forecasting, and image analysis using FastAPI, Docker, and Google Cloud services including Cloud Run and BigQuery.
- Built metadata management systems to track data flow and pipeline lineage across cloud environments, improving governance and operational visibility.

HomeGround — ML Intern | Mar 2023 – May 2023

- Developed computer vision models using PyTorch and OpenCV to detect ball-bat interactions in sports footage, improving model accuracy by 40% through feature optimization and experimentation.
- Performed dataset curation, preprocessing, and validation to improve model robustness across varying lighting and motion conditions.

PROJECTS

Flavour.AI — Multi-Agent Restaurant AI Platform

- Developed a multi-agent conversational AI platform supporting text and voice-based restaurant interactions.
- Engineered voice-enabled AI assistants using Whisper, OpenAI models, and agentic orchestration workflows.
- Designed prompt workflows for multi-turn conversations, recommendations, and structured conversational responses.

Healthcare Voice Agent – AI Appointment Booking Assistant

- Developed a voice-enabled healthcare AI assistant for appointment booking, rescheduling, cancellations, and FAQ automation.
- Engineered speech-to-text and LLM orchestration pipelines to process noisy voice inputs and generate structured appointment actions.
- Designed conversational workflows for intent detection, entity extraction, dialog state tracking, and multilingual interactions.

HR Agent ([link](#))

- Automated end-to-end recruitment workflows using a multi-agent AI system for candidate screening, evaluation, and interview coordination.
- Integrated Google Sheets, Gmail API, and Calendar APIs to automate candidate communication and interview scheduling.
- Built AI interview agents capable of conducting adaptive conversational interviews with role-specific follow-up questioning.

AI Personal Assistant & Productivity Platform – NIRVANA ([link](#))

- Built an AI-driven productivity assistant to manage tasks, habits, reminders, and workflows, Integrated to LLM wiki knowledge base and openClaw.
- Integrated Gmail and Telegram APIs for summaries, reminders, and task tracking.

Resume Optimization Agent ([link](#))

- Built an AI system to analyze, score, and optimize resumes for ATS systems using CrewAI, OpenAI, and role-specific job description analysis.

Comic App ([link](#))

- Built a social AI platform enabling collaborative comic creation, meme video generation, and AI-assisted storytelling.
- Integrated OpenAI for narrative generation and KLING/Fal.AI for image and video synthesis workflows.

ACHIEVEMENTS

- Winner of Smart India Hackathon with a ₹1,00,000 cash prize and opportunity to interact with the Prime Minister of India.
- Secured 2nd place among 3,150 teams in the AI Accenture Hackathon, winning a ₹1,00,000 cash prize.
- Finalist in the UNESCO India-Africa Hackathon, showcasing innovative AI solutions on a global platform.
- Awarded Best Customized Corpus Prize in an NLP competition among 200 participants.
- Received Best Performer Award for optimizing cost and accuracy in AI call analytics systems.

CERTIFICATIONS

- Agentic AI Course – DeepLearning.ai
- Microsoft Certified Power BI Data Analyst
- Introduction to Data Engineering
- Neo4j Certified Professional
- Claude Code Certification